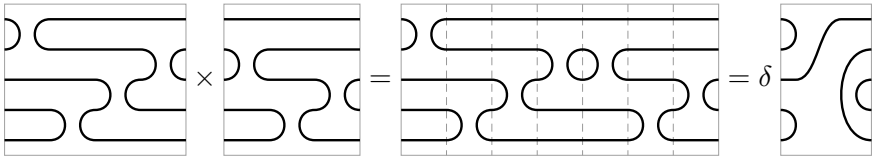
The temperleylieb package

Temperley–Lieb diagrams in LaTeX with Ti*k*Z

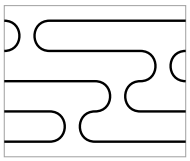
Version 1.0, 2026/10/21

The Temperley–Lieb algebra $TL_n(\delta)$ is generated by the identity $1 = e_0$ and $e_1, \dots, e_{n-1}$ subject to the Jones relations $e_i^2 = \delta e_i$, $e_i e_{i\pm1} e_i = e_i$ and $e_i e_j = e_j e_i$ for $|i-j| > 1$. Elements are drawn as diagrams of n non-crossing strands; the product is concatenation, and every closed loop is replaced by a factor δ. This package draws such diagrams from words typed exactly as you would write them in mathematics:



1 Quick start

```
\usepackage{temperleylieb}   % loads TikZ
...
\TL{5}{e_1 e_4 e_3 e_2}
```



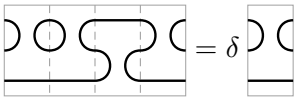
The first mandatory argument is the number n of strands, the second one is the word. A word is a sequence of generators `e_i` (or `e_{12}` for two-digit indices); `e_0` and `1` both denote the identity. Spaces, `*`, `.`, `\cdot`, `\times` and `\circ` between generators are ignored, so `e_1\cdot e_2` works as well.

2 The three features

2.1 Composition (multiplication in the algebra)

By default the generators of the word are *composed*: they are glued one after the other into a single diagram, and loops created along the way are drawn. Add `reduce` to get the reduced diagram (the basis element) with the power of δ in front; `boundaries` shows where the generators were glued.

```
\( \TL[boundaries]{3}{e_1 e_1 e_2 e_1} = \TL[reduce]{3}{e_1 e_1 e_2 e_1} \)
```

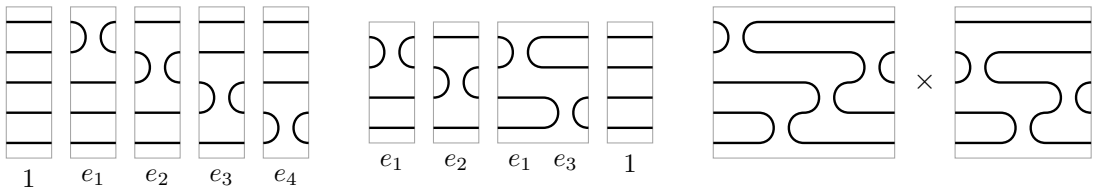


After every diagram, `\TLlastloops` expands to the number of closed loops that were found (here 1).

2.2 Displaying generators side by side without composing

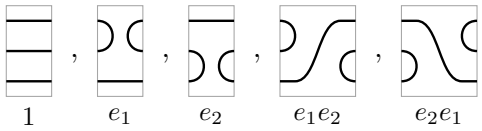
With `separate`, each generator becomes its own diagram, laid out in a row. Braces group generators that should still be composed into one factor. Alternatively, in the default `compose` mode, a bar `|` splits the word into factors. The key `separator` puts a symbol between the factors, and `names` writes the generators underneath.

```
\TLgenerators[names]{5}
\TL[separate, names]{4}{e_1 e_2 {e_1 e_3} 1}
\TL[separator=\times]{5}{e_1 e_4 e_3 e_2 | e_2 e_4 e_3}
```



`\TLgenerators[`⟨*options*⟩`]{`⟨n⟩`}` draws $1, e_1, \ldots, e_{n-1}$; the starred form omits the identity. Combining `separate` with `reduce` reduces each factor on its own, which is handy for lists of basis elements:

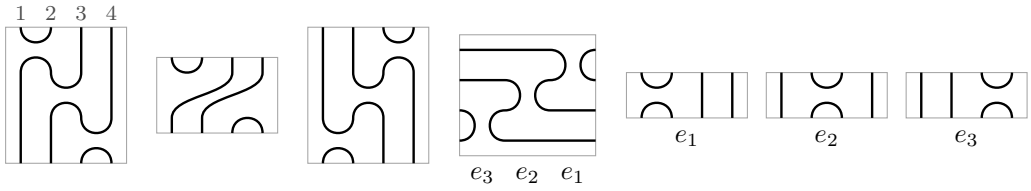
```
\TL[separate, reduce, names, separator={,}]{3}{1 e_1 e_2 {e_1 e_2} {e_2 e_1}}
```



2.3 Horizontal or vertical

Diagrams are `horizontal` by default (composition runs from left to right, strand 1 on top). With `vertical`, composition runs from top to bottom and strand 1 is on the left. `reverse` flips the direction of composition (right to left, or bottom to top).

```
\TL[vertical, labels]{4}{e_1 e_2 e_3}  \TL[vertical, reduce]{4}{e_1 e_2 e_3}
\TL[vertical, reverse]{4}{e_1 e_2 e_3}  \TL[reverse, names]{4}{e_1 e_2 e_3}
\TL[vertical, separate, names]{4}{e_1 e_2 e_3}
```



3 Reference

3.1 Commands

`\TL[`⟨***options***⟩`]{`⟨***n***⟩`}{`⟨***word***⟩`}` Draws a diagram in TL_n. Works in text and in math; the diagram is vertically centred on the math axis. `\TLdiagram` is the same command under a longer name (`\TL` is only defined if no other package has taken it).

`\TLgenerators*[`⟨***options***⟩`]{`⟨***n***⟩`}` Draws $1, e_1, \ldots, e_{n-1}$ side by side (without 1 in the starred form).

`\tlset{⟨options⟩}` Sets options for all following diagrams in the current group.

`\TLlastloops` Number of closed loops in the last diagram.

3.2 Options

Option	Default	Meaning
`compose / separate`	`compose`	glue the word, or draw each generator alone
`reduce`	false	draw reduced diagram(s) with a power of δ
`horizontal / vertical`	`horizontal`	direction of composition
`reverse`	false	compose right-to-left / bottom-to-top
`names`	false	write the generators below the diagram
`labels`	false	number the strands
`boundaries`	false	dashed lines between composed generators
`separator=`⟨*math*⟩	empty	symbol between separate factors
`coefficient`	true	print δ^k in front of reduced diagrams
`loop value=`⟨*math*⟩	`\delta`	symbol used for a loop
`box=frame|fill|none`	`frame`	background of each diagram
`strand sep=`⟨*number*⟩	0.4	distance between strands (cm)
`length=`⟨*number*⟩	0.6	length of one generator (cm)
`gap=`⟨*number*⟩	0.25	space between separate factors (cm)
`scale=`⟨*number*⟩	1	scales the picture
`baseline=`⟨*dim*⟩	`-0.5ex`	TikZ baseline (picture is centred at $y = 0$)
`strand style=`⟨*tikz*⟩		extra TikZ options for strands
`box style=`⟨*tikz*⟩		extra TikZ options for the boxes
`tikz=`⟨*tikz*⟩		extra options for the `tikzpicture`

3.3 Styles

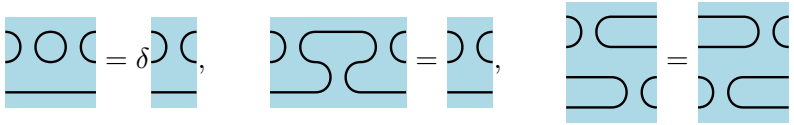
The look is controlled by TikZ styles that you can change with `\tikzset`: `tl strand`, `tl box`, `tl box frame`, `tl box fill`, `tl boundary`, `tl label`, `tl name`, `tl separator` and `tl coefficient`. The colour `tllightblue` is used by `box=fill`.

```
\tlset{box=fill, strand style={line width=1.4pt}, strand sep=0.35}
\TL[reduce, loop value=(q+q^{-1})]{4}{e_2 e_1 e_3 e_2 e_2}
\TL[box=none, strand style={red}, scale=0.7]{4}{e_2 e_1 e_3 e_2 e_2}
```

$(q + q^{-1})$ 

4 The Jones relations

```
\tlset{box=fill}
\[ \TL{3}{e_1 e_1} = \delta \TL{3}{e_1}, \qquad
   \TL{3}{e_1 e_2 e_1} = \TL{3}{e_1}, \qquad
   \TL{4}{e_1 e_3} = \TL{4}{e_3 e_1} \]
```



5 Single pictures with the standalone class

The class option `tikz` of standalone makes every Ti*k*Z picture a cropped page of its own. A diagram is part of a formula, so in such documents the package makes the formula the page instead: every `\[...\]` and `\(...\)` becomes one cropped page (this is the class option `math` of standalone, which is switched on automatically), and the diagrams inside it are ordinary parts of the formula. Diagrams outside formulas are still pages of their own.

```
\documentclass[tikz, border=1pt]{standalone}
\usepackage{amsmath,temperleylieb}
\begin{document}
\[ \TL{5}{e_1 e_4 e_3 e_2} \cdot \TL{5}{e_2 e_4 e_3}
   = \TL[reduce]{5}{e_1 e_4 e_3 e_2 e_2 e_4 e_3} \]
\end{document}
```

Other formulas, such as `$...$` or `equation*`, are not pages for standalone; a diagram in one of them gives a warning. Write the formula with `\[...\]` or `\(...\)`, or use `\documentclass[varwidth]{standalone}`, which puts the whole document on one cropped page.

6 How the reduction works

A word $e_{g_1} \cdots e_{g_m}$ is turned into a graph on the points (c, j), $0 \le c \le m$, $1 \le j \le n$: generator e_{g_k} joins column $k-1$ to column k by straight strands, except for strands g_k and $g_k + 1$, which are joined by a cup on column $k-1$ and a cap on column k. Following each strand from the boundary gives the planar matching of the reduced diagram; the remaining (internal) components are the closed loops. Reduced cups and caps are drawn with a depth proportional to their nesting level, so that nested arcs never touch.